

Assignment 01 Solution

Git, Gitea, GitHub, Git LFS, and GitHub Pages

Student: Maleeha Nadeem
Course: Cloud Computing (BSE-410)
Submission date: 10 October 2026
Total marks: 10

This document records the steps performed and the evidence collected for Assignment 01. Each task below states what was done and includes the required screenshot(s) immediately at the point they were captured.

Task 1: Install Gitea and Push a Repository from the Ubuntu Server

Step 1.1: Install and Run Gitea on the Ubuntu Server

Installed Docker Engine and curl, installed the Docker Compose CLI plugin, started the Docker service, and added my Ubuntu user to the docker group. Verified git, docker, and docker compose versions. Cloned the instructor-provided Gitea setup repository and started Gitea and PostgreSQL with docker compose up -d. Confirmed both containers were running and that Gitea responded on port 3000. Completed the Gitea web installation using the required PostgreSQL database settings.

```
* Support: https://ubuntu.com/pro

System information as of Mon Oct 5 05:45:02 AM UTC 2026

System load: 1.05          Processes: 225
Usage of /: 37.5% of 9.75GB Users logged in: 0
Memory usage: 14%         IPv4 address for ens33: 192.168.148.128
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
25 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

maleehanadeem13@ubuntu:~$ git --version; docker --version; docker compose version
git version 2.53.0
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
Docker Compose version v5.6.0
maleehanadeem13@ubuntu:~$ cd ~ && git clone https://github.com/MaqasSaleem97/Gitea.git && cd Gitea
Cloning into 'Gitea'...
remote: Enumerating objects: 16, done.
remote: Counting objects: 100% (16/16), done.
remote: Compressing objects: 100% (9/9), done.
remote: Total 16 (delta 4), reused 7 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (16/16), 5.37 KiB | 5.37 MiB/s, done.
Resolving deltas: 100% (4/4), done.
maleehanadeem13@ubuntu:~/Gitea$ docker compose up -d
[+] up 25/25
V Image postgres:17-alpine Pulled 110.4s
V Image gitea/gitea:latest Pulled 63.9s
V Network webnet Created 0.6s
V Volume gitea Created 0.2s
V Volume gitea_postgres Created 0.2s
V Container gitea Started 7.1s
V Container gitea_db Started 4.9s
maleehanadeem13@ubuntu:~/Gitea$ docker compose ps
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
gitea gitea/gitea:latest "/usr/bin/entrypoint..." gitea 25 seconds ago Up 19 seconds 22/tcp, 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
gitea_db postgres:17-alpine "docker-entrypoint.sh..." db 25 seconds ago Up 22 seconds 5432/tcp
maleehanadeem13@ubuntu:~/Gitea$ curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n" http://127.0.0.1:3000/
Gitea HTTP status: 200
maleehanadeem13@ubuntu:~/Gitea$ curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n" http://127.0.0.1:3000/
Gitea HTTP status: 200
maleehanadeem13@ubuntu:~/Gitea$
```

Step 1.2: Create and Push the Initial Repository

Created an empty public repository named Assignment01 in Gitea (no README/.gitignore/license). On the Ubuntu server, created a separate local repository, added a README.md with my full name and registration number, and committed it. Generated a Gitea personal access token (ubuntu-server-token, write:repository scope). Added the gitea remote (username only, no token in the URL) and pushed the initial commit using the token as the password when prompted.

```

maleehanadeem13@ubuntu:~$ cat > README.md << 'EOF'
> # Assignment01
>
> Name: Maleeha Nadeem
> Registration Number: FA24B1-SE-013
> EOF
maleehanadeem13@ubuntu:~$ git add README.md
maleehanadeem13@ubuntu:~$ git commit -m "Add student information"
[main (root-commit) 6717192] Add student information
 1 file changed, 4 insertions(+)
 create mode 100644 README.md
maleehanadeem13@ubuntu:~$ git remote add gitea http://maleehanadeem13@127.0.0.1:3000/maleehanadeem13/Assignment01.git
maleehanadeem13@ubuntu:~$ git push -u gitea main
Password for 'http://maleehanadeem13@127.0.0.1:3000':
remote: Failed to authenticate user
fatal: Authentication failed for 'http://127.0.0.1:3000/maleehanadeem13/Assignment01.git/'
maleehanadeem13@ubuntu:~$ git push -u gitea main
Password for 'http://maleehanadeem13@127.0.0.1:3000':
remote: Failed to authenticate user
fatal: Authentication failed for 'http://127.0.0.1:3000/maleehanadeem13/Assignment01.git/'
maleehanadeem13@ubuntu:~$ git push -u gitea main
Password for 'http://maleehanadeem13@127.0.0.1:3000':
remote: Push to create is not enabled for users.
fatal: unable to access 'http://127.0.0.1:3000/maleehanadeem13/Assignment01.git/': The requested URL returned error: 403
maleehanadeem13@ubuntu:~$ git push -u gitea main
Password for 'http://maleehanadeem13@127.0.0.1:3000':
remote: Failed to authenticate user
fatal: Authentication failed for 'http://127.0.0.1:3000/maleehanadeem13/Assignment01.git/'
maleehanadeem13@ubuntu:~$ git push -u gitea main
Password for 'http://maleehanadeem13@127.0.0.1:3000':
remote: Push to create is not enabled for users.
fatal: unable to access 'http://127.0.0.1:3000/maleehanadeem13/Assignment01.git/': The requested URL returned error: 403
maleehanadeem13@ubuntu:~$ git remote add gitea http://maleehanadeem13@127.0.0.1:3000/maleehanadeem13/Assignment01.git
error: remote gitea already exists.
maleehanadeem13@ubuntu:~$ git remote set-url gitea http://maleehanadeem13@127.0.0.1:3000/maleehanadeem13/Assignment01.git
maleehanadeem13@ubuntu:~$ git remote -v
gitea  http://maleehanadeem13@127.0.0.1:3000/maleehanadeem13/Assignment01.git (fetch)
gitea  http://maleehanadeem13@127.0.0.1:3000/maleehanadeem13/Assignment01.git (push)
maleehanadeem13@ubuntu:~$ git push -u gitea main
Password for 'http://maleehanadeem13@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 293 bytes | 58.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/maleehanadeem13/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
maleehanadeem13@ubuntu:~$ _

```

Step 1.3: Verify the Gitea Repository

Opened the Assignment01 repository in Gitea and confirmed the repository owner, repository name, that it is public, and that the rendered README.md shows my full name and registration number.

Issues

Pull Requests

Milestones

Explore

🔔

+

👤

maleehanadeem13 / Assignment01

🔗

Watching 1

☆ Star 0

🍴 Fork 0

<> Code

Issues

Pull Requests

Actions

Packages

Projects

Releases

Wiki

...

🕒 1 Commit

🌿 1 Branch

🏷️ 0 Tags

🔍 Search code...

S

Q

🌿 main

🔗

Go to file

T

Add File

<> Code

👤 maleehanadeem13

6717192996

Add student informati...

27 minutes ago

📄 README.md

Add student information

27 minutes ago

📄 README.md

✎

Assignment01

Name; Maleeha Nadeem Registration Number: FA24B1-SE-013

Description

No description provided

Manage Topics

📖 Readme

📄 24 KiB

☆ 0 Stars

👁️ 1 Watchers

🍴 0 Forks

Powered by Gitea

Version: 28.0.0

Page: 493ms

Template: 95ms

🌑 Auto

🌐 English

Licenses

API

Task 2: Push the Same Repository to GitHub

Step 2.1: Create the GitHub Repository and Verify Both Remotes

Created an empty public GitHub repository named Assignment01 (no README/.gitignore/license). Added GitHub as a second remote (github) on the same local repository used in Task 1 and verified both remotes with git remote -v, confirming no token or password appears in either URL.

```
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 293 bytes | 58.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/maleehnadeem13/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
maleehnadeem13@ubuntu:~$ cd Assignment01
maleehnadeem13@ubuntu:~/Assignment01$ git remote add github
usage: git remote add [<options>] <name> <url>

    -f, --[no-]fetch          fetch the remote branches
                              import all tags and associated objects when fetching
                              or do not fetch any tag at all (--no-tags)
    -t, --[no-]track <branch>
                              branch(es) to track
    -m, --[no-]master <branch>
                              master branch
    --[no-]mirror[=(push|fetch)]
                              set up remote as a mirror to push to or fetch from

maleehnadeem13@ubuntu:~/Assignment01$ git remote add github
usage: git remote add [<options>] <name> <url>

    -f, --[no-]fetch          fetch the remote branches
                              import all tags and associated objects when fetching
                              or do not fetch any tag at all (--no-tags)
    -t, --[no-]track <branch>
                              branch(es) to track
    -m, --[no-]master <branch>
                              master branch
    --[no-]mirror[=(push|fetch)]
                              set up remote as a mirror to push to or fetch from

maleehnadeem13@ubuntu:~/Assignment01$ https://github.com/maleehnadeem13/Assignment01.git
-bash: https://github.com/maleehnadeem13/Assignment01.git: No such file or directory
maleehnadeem13@ubuntu:~/Assignment01$ git remote add github https://github.com/maleehnadeem13/Assignment01.git
maleehnadeem13@ubuntu:~/Assignment01$ git git remote -v
git: 'git' is not a git command. See 'git --help'.

The most similar command is
  init
maleehnadeem13@ubuntu:~/Assignment01$ git remote -v
gitea  http://maleehnadeem13@127.0.0.1:3000/maleehnadeem13/Assignment01.git (fetch)
gitea  http://maleehnadeem13@127.0.0.1:3000/maleehnadeem13/Assignment01.git (push)
github https://github.com/maleehnadeem13/Assignment01.git (fetch)
github https://github.com/maleehnadeem13/Assignment01.git (push)
maleehnadeem13@ubuntu:~/Assignment01$ _
```

Step 2.2: Push to GitHub

Pushed the main branch to the github remote.

maleehanadeem13 / Assignment01

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Assignment01 Public

main 1 Branch 0 Tags

maleehanadeem13 Add student information 6717192 · 1 hour ago 1 Commit

README.md Add student information 1 hour ago

README

Assignment01

Name; Maleeha Nadeem Registration Number: FA24B1-SE-013

About

No description, website, or topics provided.

Readme Activity 0 stars 0 watching 0 forks

Releases

No releases published [Create a new release](#)

Packages

No packages published [Publish your first package](#)

Step 2.3: Verify the GitHub Repository

Opened <https://github.com/<my-github-username>/Assignment01> and confirmed the repository owner is my GitHub account, the repository name is Assignment01, the repository is public, and the rendered README shows my full name and registration number.

maleehanadeem13 / Assignment01

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Assignment01 Public

main 1 Branch 0 Tags

maleehanadeem13 Add student information 6717192 · 3 hours ago 1 Commit

README.md Add student information 3 hours ago

README

Assignment01

Name; Maleeha Nadeem Registration Number: FA24B1-SE-013

About

No description, website, or topics provided.

Readme Activity 0 stars 0 watching 0 forks

Releases

No releases published [Create a new release](#)

Packages

No packages published [Publish your first package](#)

Task 3: Track Three Large Files with Git LFS

Step 3.1: Configure Git LFS

Initialized Git LFS, confirmed its version, and configured tracking for *.bin files. Verified the resulting rule in .gitattributes.

```
maleehanadeem13@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
maleehanadeem13@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
maleehanadeem13@ubuntu:~/Assignment01$ git lfs track "*.bin"
 "*.bin" already supported
maleehanadeem13@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
maleehanadeem13@ubuntu:~/Assignment01$
```

Step 3.2: Create and Verify Three LFS Files

Created three files each larger than 100 MiB (large-file-1.bin, large-file-2.bin, large-file-3.bin) using truncate, staged .gitattributes and the three files, and verified they are tracked by Git LFS with their sizes.

```
maleehanadeem13@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
maleehanadeem13@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
maleehanadeem13@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
maleehanadeem13@ubuntu:~/Assignment01$ git add large-file-1.bin
large-file-2.bin large-file-3.bin

maleehanadeem13@ubuntu:~/Assignment01$
maleehanadeem13@ubuntu:~/Assignment01$
maleehanadeem13@ubuntu:~/Assignment01$ git lfs ls-files --size
2338a970c0 * Assignment01/large-file-2.bin (107 MB)
cb91d5908e * Assignment01/large-file-3.bin (108 MB)
f1753a034b * Assignment01/large-file-1.bin (106 MB)
maleehanadeem13@ubuntu:~/Assignment01$ git commit -m "Fix: make
```

Step 3.3: Commit and Push the LFS Files

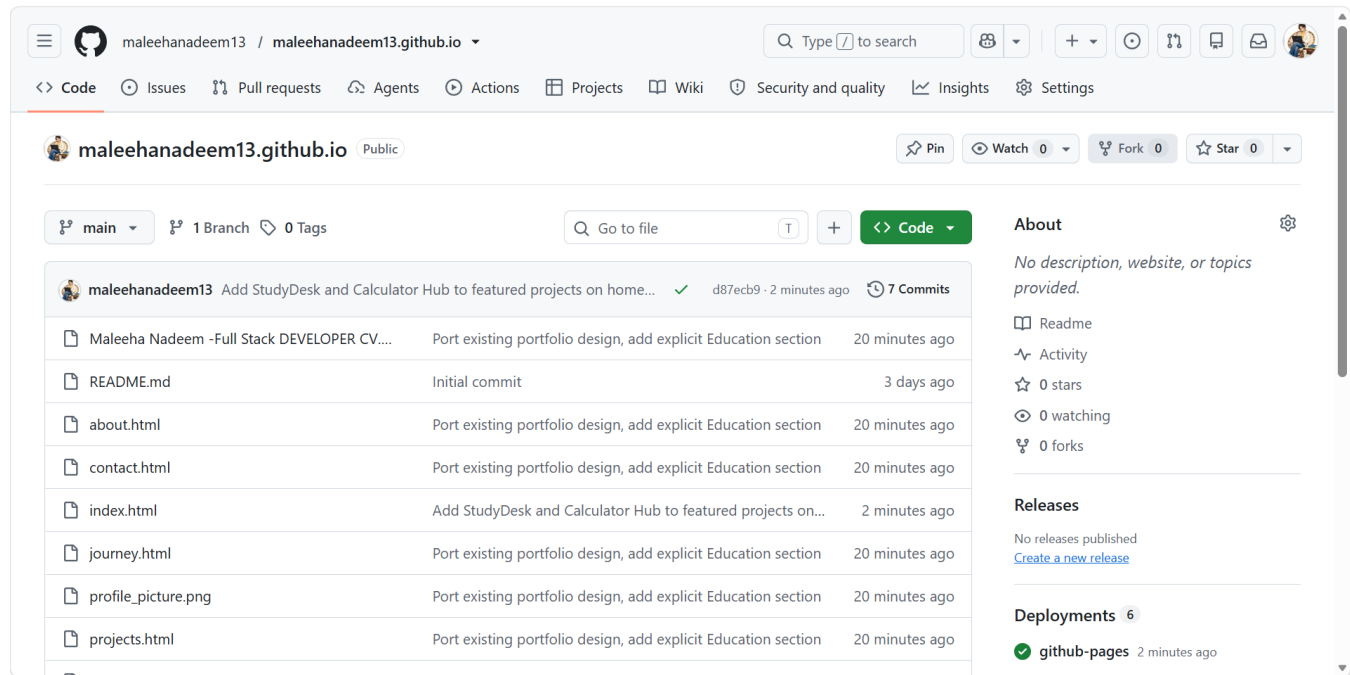
Committed the three LFS-tracked files and pushed them to the GitHub Assignment01 repository.

```
maleehanadeem13@ubuntu:~/Assignment01$ git push github main
Username for 'https://github.com': maleehanadeem13
Password for 'https://maleehanadeem13@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/maleehanadeem13/Assignment01.git/'
maleehanadeem13@ubuntu:~/Assignment01$ git push github main
Username for 'https://github.com': maleehanadeem13
Password for 'https://maleehanadeem13@github.com':
Username for 'https://github.com': maleehanadeem13
Password for 'https://maleehanadeem13@github.com':
Uploading LFS objects: 100% (2/2), 215 MB | 574 KB/s, done.
Enumerating objects: 9, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 689 bytes | 689.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/maleehanadeem13/Assignment01.git
    770c3ba..1b7e30a  main -> main
maleehanadeem13@ubuntu:~/Assignment01$
```

Task 4: Create a Portfolio or CV with GitHub Pages

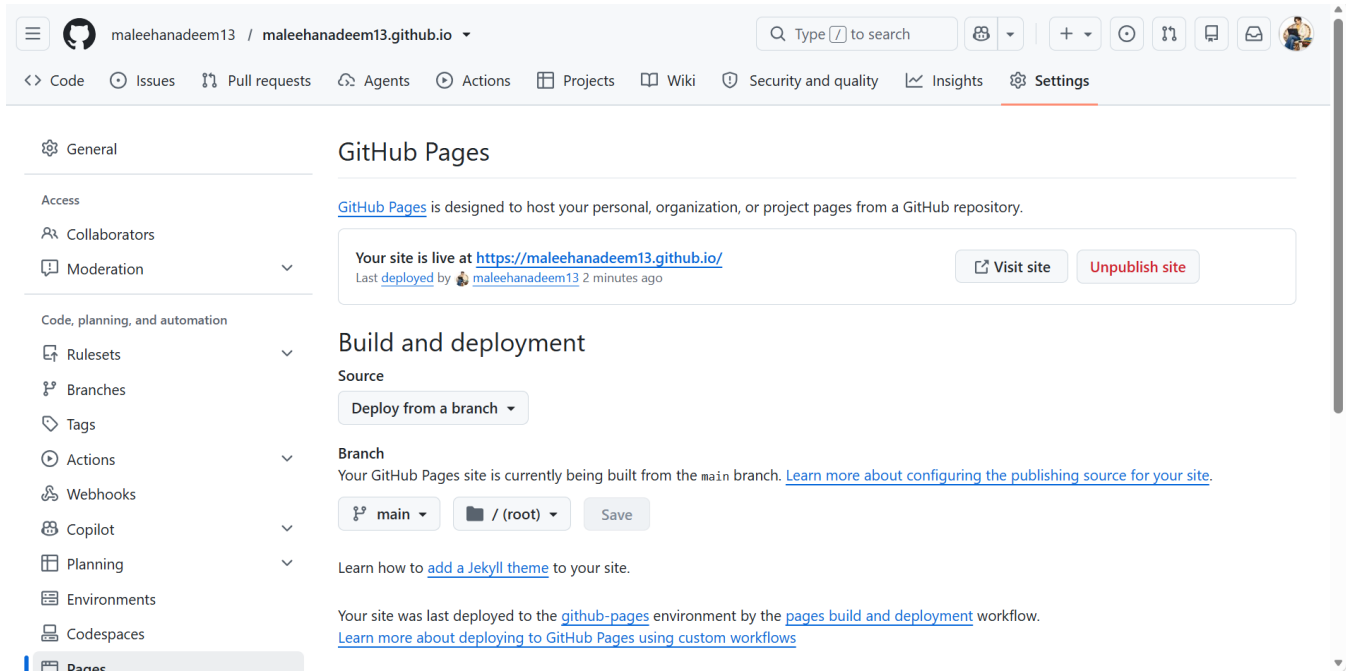
Step 4.1: Create and Push the Portfolio Repository

Created a public GitHub repository named <my-github-username>.github.io containing index.html and styles.css, with About Me, Education, Skills, and Projects sections. Committed and pushed the source files to the main branch.



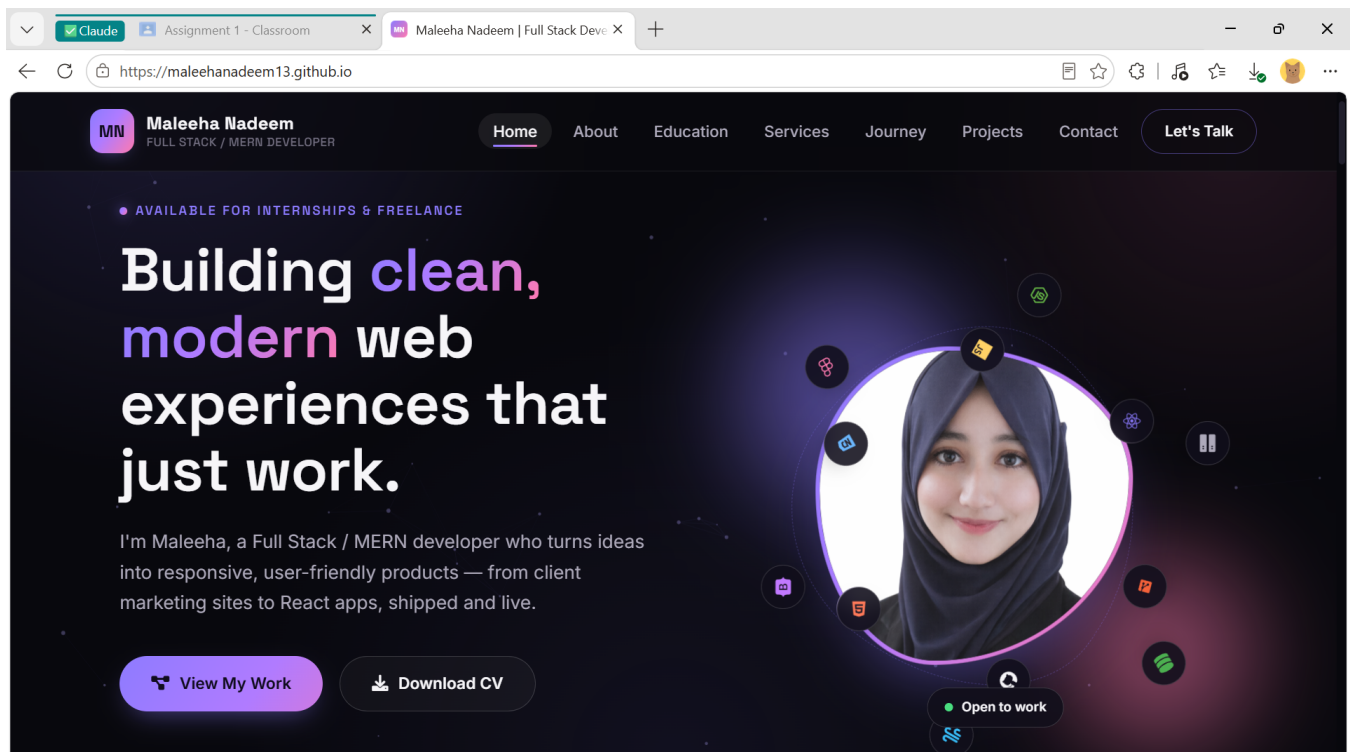
Step 4.2: Publish with GitHub Pages

In repository Settings > Pages, selected Deploy from a branch, chose the main branch and /(root) folder, saved, and waited for GitHub to report the site as published/live.



Step 4.3: Verify the Live Portfolio

Opened <https://<my-github-username>.github.io/> and confirmed the site loads successfully with the portfolio content visible.



Assignment Summary

- Installed and ran a Gitea service on an Ubuntu server.
- Performed Git work from an Ubuntu server with my GitHub identity visible.
- Pushed the same local repository to Gitea and GitHub using two remotes.
- Configured Git LFS and pushed three large files.
- Created and published a portfolio/CV with GitHub Pages.
- Prepared this Word solution and exported it as a readable PDF.

Links

Gitea repository: <http://192.168.148.128:3000/maleehanadeem13/Assignment01>

GitHub Assignment01 repository: <https://github.com/maleehanadeem13/Assignment01>

Portfolio (GitHub Pages): <https://maleehanadeem13.github.io/>

Final Submission Checklist

- I completed the assignment individually.
- Assignment01.md is inside CC/Assignments/Assignment01/.
- Assignment01_Solution.docx is inside CC/Assignments/Assignment01/ and opens correctly.
- Assignment01_Solution.pdf is inside CC/Assignments/Assignment01/, matches the Word solution, and opens correctly.
- My CC repository contains Assignments/Assignment01/screenshots/.
- All 12 required screenshots are present and readable.
- Every screenshot was captured immediately after the requested step.
- Every screenshot was created from my own work.
- My GitHub username is visible in every Ubuntu server terminal screenshot.
- Every terminal screenshot shows the required command and its relevant result.
- Every browser screenshot shows the required repository, URL, page, or status.
- No screenshot contains a token, password, private key, or other credential.
- My CC, GitHub Assignment01, and <username>.github.io repositories are public.
- My Gitea Assignment01 repository was created and pushed from the Ubuntu server.
- All repositories use the main branch.
- I pushed the complete Assignments/Assignment01 directory before the deadline.
- I uploaded all three required assignment files to Google Classroom.
- I submitted the three required links on Google Classroom.